
Discovering Cooperative Pipelines: Autoresearch for Sequential Social Dilemmas

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Abstract

1 We study *two-level autoresearch* for cooperation: an outer-loop AI agent au-
2 tonomously redesigns the inner-loop pipeline of an LLM policy-synthesis sys-
3 tem for multi-agent Sequential Social Dilemmas (SSDs). A *researcher agent* $\mathcal{R}$
4 (run as a coding agent) reads the inner-loop source code, edits system prompts,
5 feedback functions, helper libraries, and iteration logic, runs evaluations, and
6 decides what to keep, following the *autoresearch* paradigm. Across two games
7 (Cleanup and Gathering), two policy-synthesizer LLMs, and two welfare objec-
8 tives (utilitarian efficiency and Rawlsian maximin), the researcher reliably exceeds
9 hand-designed baselines, sharply tightens run-to-run variance, and outperforms
10 prompt-only optimization. The discovered pipelines are objective-dependent: un-
11 der maximin the researcher independently rediscovers *duty rotation* as a fairness
12 mechanism, supporting an *information-design* reading in which the researcher
13 chooses what to reveal to the boundedly rational synthesizer. Code at <https://github.com/anon590/llm-policies-social-dilemmas>.
14

15 1 Introduction

16 Sequential Social Dilemmas (SSDs) [1] are the multi-agent analogue of the prisoner’s dilemma in
17 temporally rich Markov games: individually rational play leads to collectively suboptimal outcomes
18 through pollution, over-harvesting, or open conflict. Standard multi-agent reinforcement learning
19 (MARL) struggles in this regime due to credit assignment, non-stationarity, and large joint action
20 spaces [2]. A complementary approach, recently introduced by Gallego [3], replaces parameter-
21 space optimization with *algorithm-space synthesis*: a frozen LLM writes a Python policy function,
22 evaluates it in self-play, and iteratively refines it from performance feedback. A single generation
23 step can produce coordination logic (territory partitioning, role assignment, conditional cooperation)
24 at a sample efficiency several orders of magnitude beyond what gradient-based MARL achieves on
25 the same environments.

26 This shifts where the design problem lives, rather than removing it. The inner-loop pipeline that
27 drives the synthesizer has many free parameters: which system prompt, which feedback variables,
28 which helper functions, how many refinement steps. Each materially affects the resulting policies,
29 and prior work tuned them by hand. A natural question follows: *can an AI agent design the pipeline?*

30 We answer affirmatively with a two-level autoresearch framework. An *outer-loop researcher agent*
31 (Claude Opus 4.6, run as a coding agent) edits the source files of an *inner-loop policy synthesizer*
32 (another LLM), runs evaluations on held-out seeds, and keeps modifications that improve a fixed
33 welfare objective Φ . The outer agent operates on an ordinary git repository (reading code, writing
34 diffs, running shell commands, etc) without task-specific scaffolding, mirroring the autoresearch
35 paradigm of Karpathy [4] for single-GPU LLM pretraining. Although the inner-loop SSDs are
36 gridworld benchmarks rather than physical systems, the outer-loop discovery process itself runs

under conditions a deployed discovery agent faces: noisy multi-seed evaluations, stochastic code generation, an LLM-evaluation budget that bounds how often Φ can be queried, and a heterogeneous code repository the agent must navigate end-to-end on its own.

Our contributions are: i) a general two-level framework that delegates the design of an LLM synthesis pipeline to a coding agent operating on a real software repository (Section 3); ii) the first instantiation of the autoresearch paradigm in a multi-agent decision-making domain, with experiments across two SSDs (Cleanup and Gathering), two policy LLMs, and two welfare objectives (utilitarian efficiency U and Rawlsian maximin $\min_i R_i$) (Section 4); and iii) a mechanism-design interpretation supported by the qualitatively different pipelines the agent produces under different welfare objectives, including the autonomous discovery of *time-based duty rotation* as a fairness mechanism, never found under efficiency optimization.

2 Background

We build on the iterative LLM policy synthesis framework of Gallego [3], which serves as the (frozen) inner loop of our two-level system. This section recalls the SSD formalism, the social outcome metrics, and the base synthesis loop.

2.1 Sequential Social Dilemmas

A Sequential Social Dilemma is a partially observable Markov game $\mathcal{G} = \langle N, \mathcal{S}, \{\mathcal{A}_i\}_{i=1}^N, T, \{R_i\}_{i=1}^N, H \rangle$ with N agents, state space $\mathcal{S}$ (the gridworld configuration), per-agent action spaces $\mathcal{A}_i$, transition function T , reward functions R_i , and episode horizon H [1]. Beyond the dilemma’s matrix-game structure, SSDs add temporal richness: agents must learn *when* and *where* to cooperate, not just *whether* to.

We study two canonical SSDs that capture complementary dilemma types.

Cleanup [5] is a *public goods provision* game ($N=10$). The map has two regions: a river that accumulates waste, and an orchard where apples grow. Apples regrow only when river pollution is below threshold. Each agent can fire a cleaning beam (cost -1) that removes waste, collect apples ($+1$ each), or fire a tagging beam (cost -1 , inflicting -50 on the target and removing it for 25 steps). The dilemma: cleaning is costly but its benefits are public, so purely self-interested agents free-ride.

Gathering [1, 6] is a *common pool resource* game ($N=4$). Agents collect shared apples on a fixed respawn timer and may fire tagging beams to temporarily remove rivals. The dilemma: agents can coexist and share resources, or aggress to monopolize them; aggression wastes time and reduces total welfare.

Both games use 8–9 discrete actions (movement, rotation, beam, stand, optionally clean) and episodes of $H=1000$ steps. The two dilemmas differ in cost structure: asymmetric provision (cleaners pay, all benefit) vs. symmetric restraint (every agent faces the same temptation), a distinction that drives our experimental findings.

Social metrics. Following Perolat et al. [6], let $R_i = \sum_{t=0}^{H-1} r_i^t$ denote agent i ’s episode return. We evaluate four social outcomes:

$$U = \frac{1}{H} \sum_{i=1}^N R_i \quad (\text{efficiency}) \quad (1)$$

$$E = 1 - \frac{\sum_{i,j} |R_i - R_j|}{2N \sum_i R_i} \quad (\text{equality}) \quad (2)$$

$$S = \frac{1}{N} \sum_{i=1}^N \bar{t}_i \quad (\text{sustainability}) \quad (3)$$

$$P = \frac{1}{H} \sum_{t=0}^{H-1} |\{i : \text{active}_i^t\}| \quad (\text{peace}) \quad (4)$$

where $\bar{t}_i$ is the mean timestep at which agent i collects positive reward (higher $\Rightarrow$ resources preserved later) and active_i^t indicates that agent i is not tagged out at step t . We additionally consider the

76 *maximin* (Rawlsian) welfare criterion $\min_i R_i$, which optimizes the worst-off agent’s return and
 77 serves as the second objective in our experiments.

78 2.2 Iterative LLM Policy Synthesis

79 Let Π denote the space of *code-based policies*: deterministic functions $\pi : \mathcal{S} \times [N] \rightarrow \mathcal{A}$ expressed
 80 as executable Python code. Each policy has access to the full environment state and a library of
 81 helpers (BFS pathfinding, beam targeting, coordinate transforms). This state access is a deliberate
 82 design choice: programmatic policies operate in algorithm space rather than the reactive observation-
 83 to-action space of neural policies, which lets a single LLM generation step encode rich coordination
 84 logic.

85 A frozen LLM $\mathcal{M}$ acts as the *policy synthesizer*. Given a system prompt p describing the environ-
 86 ment API and a feedback prompt q_k , it produces a new policy

$$\pi_{k+1} = \mathcal{M}(p, q(\pi_k, \mathcal{F}_k)),$$

87 where π_k is the previous policy (its source code) and $\mathcal{F}_k$ is the evaluation feedback. All N agents
 88 execute the same program π_k in self-play. We stress that this is *symmetric*, not *behaviorally ho-*
 89 *mogeneous*: since π_k takes `agent_id` as an argument, a single shared program can induce distinct
 90 per-agent behaviors (cleaner vs. gatherer assignment, time-rotated duty cycles, partitioned territo-
 91 ries; see the synthesized policies in Appendix E). What is shared is the source code; the sampled
 92 action distributions can differ across agents. Evaluation over a set of random seeds S yields the
 93 mean per-agent return $\bar{r}_k$ and the social metrics vector $\mathbf{m}_k = (U_k, E_k, S_k, P_k)$. Each generated
 94 policy passes an AST-based safety check (blocking eval, file I/O, network access) followed by a
 95 short smoke test; failures trigger regeneration (up to R attempts) with the error message appended
 96 to the prompt.

97 **Feedback.** We package the previous policy’s source code together with all available evaluation
 98 signals:

$$\mathcal{F}_k = (\text{code}(\pi_k), \bar{r}_k, \mathbf{m}_k, \mathbf{d}), \quad (5)$$

99 where $\mathbf{d}$ contains natural-language definitions of each social metric. The LLM consumes $\mathcal{F}_k$ to
 100 revise and improve the policy. This is a starting point; the choice of feedback content is a single
 101 design decision within a much larger pipeline configuration space: our two-level framework (Sec-
 102 tion 3) opens the full space to automated search.

103 3 Two-Level Framework

104 We introduce a two-level system where a *researcher agent* $\mathcal{R}$ autonomously discovers configurations
 105 that optimize the output of an inner-loop system. While we instantiate this for multi-agent policy
 106 synthesis, the architecture is general: any pipeline where an LLM generates artifacts, evaluates
 107 them, and iterates can serve as the inner loop. The fundamental insight is that the entire inner-loop
 108 codebase is a designable artifact that a code-based agent can search over. Figure 1 illustrates the
 109 architecture.

110 3.1 Configuration Space

111 Let $\mathcal{C}$ denote the space of *pipeline configurations*. Each configuration $c \in \mathcal{C}$ specifies the full inner-
 112 loop setup:

$$c = (p, \phi, \mathcal{H}, \iota) \quad (6)$$

113 where p is the system prompt, ϕ is the feedback construction function (which metrics and diagnostics
 114 to include, how to frame it, whether to inject adaptive hints and thresholds, etc), $\mathcal{H}$ is the helper func-
 115 tion library (auxiliary functions for pathfinding, getting aggregates of useful environment quantities,
 116 etc.), and ι specifies the iteration logic (number of inner iterations K , sampling strategy). Table 1
 117 provides concrete examples. The validation pipeline is part of the frozen inner-loop infrastructure
 118 rather than a configurable component, the researcher cannot modify it in our experiments to prevent
 119 reward hacking.

120 The hand-designed feedback of [3] corresponds to a single fixed instantiation of ϕ . Our framework
 121 opens the full configuration space to automated search.

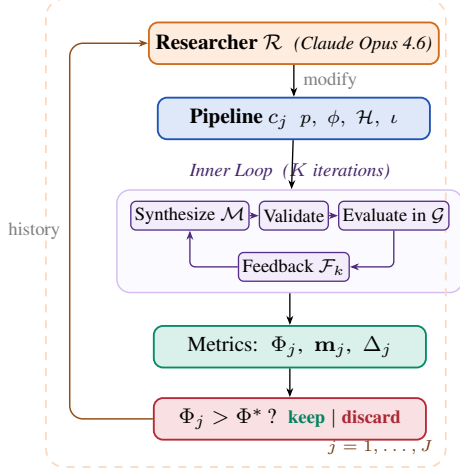


Figure 1: Two-level automated research framework (Algorithm 1).

Algorithm 1 Two-Level Automated Research

Require: Game $\mathcal{G}$, policy synthesizer $\mathcal{M}$, researcher $\mathcal{R}$, system prompt $p_{\mathcal{R}}$, initial config c_0 , outer iterations J , ground-truth objective Φ , held-out seeds S_{ho}

Ensure: Best configuration c^* , best policy π^*

```

1:  $\pi_0^* \leftarrow \text{INNERLOOP}(\mathcal{M}, \mathcal{G}, c_0)$ 
2:  $\Phi_0 \leftarrow \text{EVAL}(\pi_0^*; \mathcal{G}, S_{\text{ho}})$ 
3:  $\text{history} \leftarrow \{(c_0, \Phi_0, \mathbf{m}_0, \emptyset)\}$ 
4: for  $j = 1, \dots, J$  do
5:    $c_j \leftarrow \mathcal{R}(p_{\mathcal{R}}, \text{CODE}(c_{j-1}), \text{history})$ 
6:   if  $\neg \text{VALIDATECONFIG}(c_j)$  then  $\text{retry} \leq R$ 
7:      $\pi_j^* \leftarrow \text{INNERLOOP}(\mathcal{M}, \mathcal{G}, c_j)$ 
8:      $\Phi_j \leftarrow \text{EVAL}(\pi_j^*; \mathcal{G}, S_{\text{ho}})$ 
9:      $\Delta_j \leftarrow \text{DIFF}(c_{j-1}, c_j)$  // code diff
10:     $\text{history} \leftarrow \text{history} \cup \{(c_j, \Phi_j, \mathbf{m}_j, \Delta_j)\}$ 
11:  end for
12:  $j^* \leftarrow \arg \max_j \Phi_j$ 
13: return  $c_{j^*}, \pi_{j^*}^*$ 

```

Table 1: Configuration components modifiable by the researcher, with concrete edits that $\mathcal{R}$ made in our runs. The environment simulator, ground-truth evaluation, and policy LLM weights are frozen.

	Scope	Concrete edits by $\mathcal{R}$ (see appendix)
p	System prompt	Multi-section strategic briefing with explicit duty-rotation template (agent_id + step//T) % n (App. D.2, Listing 3)
ϕ	Feedback fn	Thresholded diagnostics: “FAIRNESS ALERT” fires when $\min_i R_i < 0$; “DO NOT REGRESS” guard fires when $U \geq 2.5$ (App. D.3, Listing 4)
$\mathcal{H}$	Helper library	BFS-Voronoi territories with respawn-timer awareness; band-based apple zoning by agent_id (App. D.1, Listings 2, 1)
ι	Iteration logic	Per-condition $K \in \{2, 3\}$; $ S $ walked 5→8→12 for variance control; thinking budget 10–32k (App. D.4, Table 6)

3.2 Inner Loop (Policy Synthesis)

Given a configuration c , the inner loop executes K iterations of LLM policy synthesis:

$$\pi_c^* = \text{INNERLOOP}(\mathcal{M}, \mathcal{G}, c) \quad (7)$$

Each iteration k proceeds in four stages, following [3]:

- Synthesize.** The policy LLM $\mathcal{M}$ receives the system prompt p , the previous policy’s source code π_{k-1} , and feedback $\mathcal{F}_{k-1}$ constructed by ϕ . It generates a new Python policy function π_k that has access to full environment state and the helper library $\mathcal{H}$.
- Validate.** The generated code undergoes AST-based safety checks (blocking dangerous operations such as file I/O and network access) followed by a short smoke test. Failures trigger re-generation (up to R retries), with the error message appended to the prompt.
- Evaluate.** All N agents execute the same policy π_k in self-play over $|S|$ random seeds (note the policy is conditional on agent_id). The evaluation yields the mean per-agent reward $\bar{r}_k$ and the social metrics vector $\mathbf{m}_k = (U_k, E_k, S_k, P_k)$.
- Feedback.** The feedback function ϕ constructs the prompt for the next iteration from $(\pi_k, \bar{r}_k, \mathbf{m}_k)$, packaging the previous policy’s source code together with the scalar reward, the social metrics vector, their natural-language definitions, and any adaptive diagnostics that ϕ injects.

The inner loop output is evaluated on held-out seeds via a fixed ground-truth objective:

$$\Phi(c) = \text{EVAL}(\pi_c^*; \mathcal{G}, S_{\text{held-out}}) \quad (8)$$

139 where Φ maps from social outcomes to a scalar. We consider two alternative objectives to maximize:

$$\Phi_U = U = \frac{1}{H} \sum_{i=1}^N R_i \quad (\text{utilitarian efficiency}) \quad (9)$$

$$\Phi_{\min} = \min_i R_i \quad (\text{Rawlsian maximin}) \quad (10)$$

140 Φ_U rewards collective throughput and is indifferent to how reward is distributed across agents,
 141 whereas $\Phi_{\min}$ instead optimizes for the worst-off agent, pressuring the researcher toward configura-
 142 tions that distribute the cost of cooperation.

143 3.3 Outer Loop (Automated Research)

144 The researcher agent $\mathcal{R}$ iteratively modifies the pipeline configuration. Following the autoresearch
 145 paradigm [4], $\mathcal{R}$ operates on the inner-loop codebase as a modifiable artifact, proposing changes,
 146 observing outcomes, and refining. The procedure is formalized in Algorithm 1.

147 At each outer iteration j , the researcher $\mathcal{R}$ receives: i) the **full source code** of the current config-
 148 uration c_{j-1} (prompts, feedback construction, helpers, iteration logic); ii) the **experiment history**:
 149 for each prior iteration, the code diff Δ_i , ground-truth score Φ_i , and social metrics vector $\mathbf{m}_i$; iii)
 150 the **environment source code** (read-only), enabling the researcher to reason about game mechanics.
 151 The researcher proposes a new configuration c_j by generating code modifications. Concretely, $\mathcal{R}$
 152 is a coding agent (Claude Code CLI) that operates on a real software repository: it reads and edits
 153 Python source files, runs shell commands, inspects evaluation outputs, and commits changes to a
 154 dedicated git branch, following the same workflow a human researcher would follow.

155 3.4 Connection to Automated Mechanism Design

156 The two-level structure admits a mechanism design interpretation. The researcher $\mathcal{R}$ acts as a *mech-*
 157 *anism designer*: it controls the information structure (what metrics to reveal, how to frame them),
 158 the action space (what helper functions are available), and the incentive structure (how feedback is
 159 presented) under which the policy synthesizer $\mathcal{M}$ operates. The synthesizer acts as the *agent* within
 160 the designed mechanism.

161 This connects to the automated mechanism design literature [7], where a principal designs rules to
 162 induce desired behavior from self-interested agents. In our setting:

- 163 • The **principal** is the researcher $\mathcal{R}$, optimizing the ground-truth objective Φ .
- 164 • The **agent** is the synthesizer $\mathcal{M}$, optimizing per-agent reward as instructed.
- 165 • The **mechanism** is the configuration c : prompts, feedback, helpers, iteration logic.
- 166 • The **outcome** is the social welfare of the resulting multi-agent policy π_c^* .

167 A crucial distinction from classical mechanism design: $\mathcal{M}$ follows instructions but has *bounded*
 168 *rationality* in the sense that its ability to synthesize effective policies depends on the information
 169 and tools provided. The researcher’s task is thus closer to *information design* [8]: choosing what
 170 to reveal to help $\mathcal{M}$ navigate the cooperation–defection tension. Our experiments (Section 4) show
 171 that the researcher designs qualitatively different information structures depending on the welfare
 172 objective Φ , supporting this interpretation empirically.

173 4 Experiments

174 4.1 Setup

175 We conduct 12 autonomous researcher runs across a factorial design. The researcher agent $\mathcal{R}$ is
 176 Claude Opus 4.6, invoked via the Claude Code CLI as a coding agent. Each run operates on a dedi-
 177 cated git branch of a real Python codebase: $\mathcal{R}$ edits source files in `pipeline/`, executes evaluation
 178 scripts, reads metric outputs, and iterates, without human intervention.

179 **Design.** For Cleanup: 2 policy LLMs $\times$ 2 objectives $\times$ 2 replications = 8 runs. For Gathering: 2
 180 policy LLMs $\times$ 1 objective $\times$ 2 replications = 4 runs. Maximin runs are unnecessary for Gathering
 181 because efficiency optimization alone achieves close to perfect equality.

Table 2: Results. Mean / max across the runs per condition. **Bold** marks the best mean per metric within each Policy LLM $\times$ Game sub-block. Baseline: unmodified, hand-designed pipeline from [3]. GEPA: automated prompt optimization [9] with matched compute budget.

Policy LLM	Target	U	E	$\min_i R_i$
<i>Cleanup — Baseline</i>				
Gemini 3.1 Pro	—	1.93/2.70	0.17/0.62	−159/−84
Sonnet 4.6	—	0.86/1.56	−0.02/0.25	−151/−59
<i>Cleanup — GEPA [9]</i>				
Gemini 3.1 Pro	Φ_U	1.34/1.37	−0.10/−0.05	−149/−126
	$\Phi_{\min}$	1.76/2.76	0.90/0.96	143/245
Sonnet 4.6	Φ_U	1.04/1.20	−0.59/−0.21	−164/−126
	$\Phi_{\min}$	−0.63/1.60	0.77/1.00	−147/6
<i>Cleanup — Two-level Autoresearch</i>				
Gemini 3.1 Pro	Φ_U	3.20 /3.25	0.55/0.61	−211/−182
	$\Phi_{\min}$	3.16/3.19	0.98 /0.98	290 /296
Sonnet 4.6	Φ_U	3.12 /3.14	0.66/0.70	−196/−146
	$\Phi_{\min}$	2.57/2.93	0.91 /0.97	179 /200
<i>Gathering — Baseline</i>				
Gemini 3.1 Pro	—	2.04/2.42	0.90/0.98	412/571
Sonnet 4.6	—	0.03/0.03	0.54/0.54	0/0
<i>Gathering — GEPA [9]</i>				
Gemini 3.1 Pro	Φ_U	2.08/2.35	0.94/0.96	436/518
Sonnet 4.6	Φ_U	1.20/1.23	0.63/0.71	86/123
<i>Gathering — Two-level Autoresearch</i>				
Gemini 3.1 Pro	Φ_U	2.49 /2.51	0.98 /0.98	582 /582
Sonnet 4.6	Φ_U	2.52 /2.52	0.96 /0.98	576 /597

Models. Policy synthesizer $\mathcal{M}$: Gemini 3.1 Pro (Google) or Claude Sonnet 4.6 (Anthropic), to recent state-of-the-art LLMs. Both use extended thinking. We additionally test with Gemma 4 26B-A4B-IT (Google), a smaller open-weight model, to probe the framework’s behavior when the policy synthesizer has substantially lower capability (Appendix C).

Baselines. The hand-designed feedback configuration from prior work [3] serves as the initial pipeline c_0 for all runs. On Cleanup ($N=10$), this baseline achieves $U=1.93/2.70$ (mean/max) with Gemini and $U=0.86/1.56$ with Sonnet. We additionally compare against GEPA [9], an automated prompt optimization method that iteratively refines the system prompt via LLM reflection. GEPA optimizes only the system prompt p , whereas our researcher modifies the full pipeline $(p, \phi, \mathcal{H}, \iota)$. We give GEPA a matched compute budget: the same number of optimization steps as outer iterations used by our method, so all in all both methods use a comparable number of environment evaluations.

4.2 Results

Table 2 presents the main results, comparing our autoresearch framework against the hand-designed baseline of [3] and GEPA [9].

Finding 1: The researcher reliably improves over hand-designed baselines and outperforms prompt-only optimization. All 12 runs show substantial improvement regardless of starting point (Figure 2). In Cleanup, the researcher exceeds the unmodified pipeline baseline by 66% with Gemini ($U: 1.93 \rightarrow 3.20$) and 263% with Sonnet ($U: 0.86 \rightarrow 3.12$). Strikingly, the Sonnet improvement nearly closes the gap with Gemini ($U=3.12$ vs. 3.20), despite a $2\times$ baseline disadvantage (0.86 vs. 1.93). Final efficiency values cluster tightly around $U \approx 3.1$ –3.2 despite baselines varying from 0.29 to 2.70 across individual runs, suggesting the researcher reliably discovers the performance ceiling of each policy LLM. Beyond improving the mean, the researcher yields remarkably consistent results: under efficiency optimization, Gemini’s runs cluster within $U=3.20$ –3.25 (gap 0.05) despite baselines spanning 1.15–2.70, and Sonnet’s within 3.12–3.14 (gap 0.02) from baselines of 0.38–1.56. In Gathering, all four runs converge to $U=2.47$ –2.52 from baselines ranging 0.03–

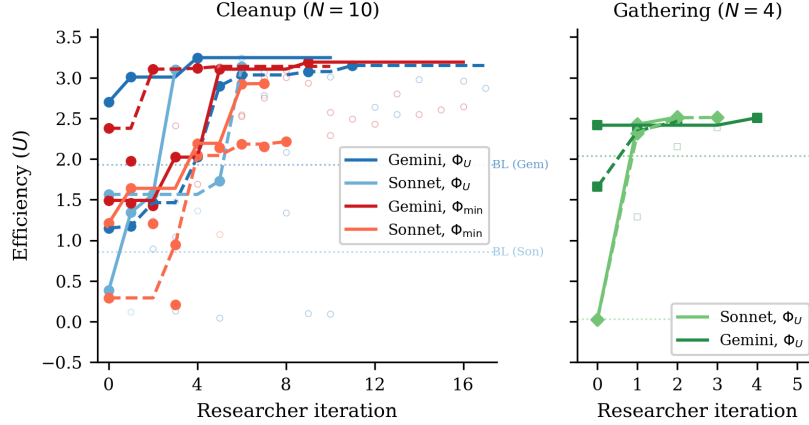


Figure 2: Efficiency (U) across researcher iterations for all 12 runs. *Left*: Cleanup ($N=10$) with 8 runs across 2 LLMs $\times$ 2 objectives (solid lines = kept experiments, open circles = discarded). Dashed horizontal line: hand-designed baseline from [3]. All runs converge to $U \approx 3.1$ – 3.2 despite diverse starting points. *Right*: Gathering ($N=4$) with 4 runs (2 per LLM). All converge to $U \approx 2.5$ within 2–4 iterations despite baselines ranging from 0.03 to 2.42.

207 2.42. The researcher discovers pipeline modifications: structured helpers (App. D.1), strategic hints
 208 (App. D.2), tailored feedback (App. D.3). These constrain the policy LLM toward consistently good
 209 solutions.

210 Compared to GEPA (Table 2), autoresearch achieves $2.4\times$ higher efficiency under Φ_U in Cleanup
 211 ($U=3.20$ vs. 1.34), $2\times$ higher maximin under $\Phi_{\min}$ (290 vs. 143), and 20% higher efficiency in Gath-
 212 ering ($U=2.49$ vs. 2.08). Critically, GEPA shows far wider spread across runs on the harder fairness
 213 objective ($\Phi_{\min}$ maximin in Cleanup: $245/41$ vs. autoresearch’s $296/284$). The gap widens further
 214 with Sonnet: on Cleanup Φ_U , GEPA stalls at $U=1.04$ ($3\times$ below autoresearch’s 3.12), and on $\Phi_{\min}$
 215 its two runs split between a modestly positive solution ($\min_i R_i=6$, $U=1.60$) and a pathological “ev-
 216 eryone cleans, nobody eats” regime (perfect equality $E=1.00$ but negative efficiency $U=-2.87$ and
 217 $\min_i R_i=-299$), while autoresearch with the same model reaches $\min_i R_i=179/200$ reliably. This
 218 highlights the advantage of modifying the full pipeline, whereas prompt-only optimization cannot
 219 address the underlying stochasticity of code generation.

220 **Finding 2: No efficiency–fairness tradeoff in Cleanup (Gemini).** Maximin-optimized Gem-
 221 ini pipelines sacrifice only 1% efficiency (U : 3.16 vs. 3.20) while achieving near-perfect equal-
 222 ity (E : 0.98 vs. 0.55) and transforming maximin from deeply negative baselines (-99 to -84) to
 223 $\min_i R_i = 290$. The researcher discovers *fair duty rotation* (Listing 6; primed by the prompt rewrite
 224 of Listing 3)—time-based cycling using `agent_id` and `env._step_count`—simultaneously im-
 225 proving worst-off welfare and collective output. Because cleaning is a public good, distributing the
 226 cleaning cost fairly ensures enough cleaners to sustain apple production.

227 For Sonnet, there is a moderate tradeoff: efficiency drops from 3.12 to 2.57 under maximin opti-
 228 mization. The gap reflects Sonnet’s harder time implementing complex coordination mechanisms
 229 (role rotation, zone assignment) from strategic hints alone.

230 **Finding 3: Game structure determines whether fairness requires explicit optimization.** In
 231 Cleanup, where cleaning costs are borne asymmetrically (cleaners pay -1 , free-riders collect ap-
 232 ples), baseline equality ranges from $E=0.04$ to 0.62 , and maximin optimization is required to reach
 233 $E > 0.9$. In Gathering, where all agents face a symmetric landscape, efficiency optimization alone
 234 achieves $E > 0.94$ across all 4 runs: no separate maximin runs are needed.

235 This generalizes: in *provision* dilemmas with asymmetric costs, fairness requires designed mecha-
 236 nisms (role rotation, duty sharing); in *restraint* dilemmas with symmetric costs, fairness emerges as a
 237 free byproduct of efficient coordination. The researcher independently discovers this, it creates role
 238 differentiation pipelines only for Cleanup, and pure spatial-coordination pipelines for Gathering.

Finding 4: Convergent discovery of qualitatively different strategies per objective. Despite fully independent runs, the researcher converges on the same core strategies within each condition (Table 3, Appendix A). In Cleanup, waste-counting helpers and spatial zone partitioning appear across all 8 runs. The key qualitative difference is *role rotation*, appearing in 4/4 maximin runs (Listings 6, 7) but 0/4 efficiency runs (Listing 5). Under efficiency optimization, the researcher assigns static cleaning roles (some agents always clean), achieving high collective output at the cost of equality. Under maximin optimization, it discovers rotation as a fairness mechanism.

In Gathering, the researcher discovers BFS-Voronoi territory partitioning and respawn-timer awareness (Listings 2, 8), with no role differentiation. Thus, optimization is purely spatial (who collects which apples) and temporal (respawn-aware positioning).

Common failure modes. Several patterns led to discarded experiments across all runs: (1) *over-prescription*: adding too many strategic hints confuses the policy LLM, causing generation failures; (2) *iteration regression*: $K=4$ or $K=5$ inner iterations often cause catastrophic regression as the LLM “over-refines” a working policy; (3) *feedback overload*: verbose per-agent diagnostics cause over-correction rather than gradual improvement; (4) *variance exposure*: identical configurations can yield $U=3.2$ then $U=0.09$ on different seeds, requiring multi-seed evaluation for stable signals (per-run $K, |S|$ in Table 6).

256 5 Related Work

Sequential social dilemmas. SSDs were introduced by Leibo et al. [1] (Gathering) and extended to public-goods settings by Hughes et al. [5] (Cleanup); Perolat et al. [6] formalized the social outcome metrics (U, E, S, P) used here. Subsequent work studies inequity aversion, intrinsic motivation, and reputational mechanisms for promoting cooperation in MARL agents. We complement this line by automating the search for cooperative *programs* rather than evolving neural policies, and by treating the welfare objective Φ as a designable parameter that the outer agent optimizes for, obtaining qualitatively different cooperative behaviors (static role assignment vs. duty rotation) under different objectives without changing the environment.

LLMs for policy and program synthesis. FunSearch [10] evolves programs for combinatorial discovery; Eureka [11] synthesizes reward functions from environment source code; Voyager [12] and Code as Policies [13] generate executable skill code for single-agent embodied control; ReEvo [14] evolves heuristics with reflective feedback. These works target single-agent settings or non-strategic optimization. Gallego [3], on which our inner loop is based, extended LLM program synthesis to the multi-agent SSD setting, where one program must coordinate N self-play copies. Our contribution is one level above: rather than tuning the synthesizer for one task, we let an outer agent rewrite the synthesis pipeline itself.

LLM reflection and prompt optimization. Reflexion [15], Self-Refine [16], OPRO [17], and GEPA [9] demonstrate that structured verbal feedback loops improve LLM outputs; ERL [18] internalizes such reflection via self-distillation. These methods optimize the prompt or the model’s reasoning trajectory in isolation. Our framework instead modifies the entire surrounding pipeline (system prompt, feedback construction, helper library, iteration logic) making prompt optimization one component of a strictly larger search space. The empirical gap is large: Section 4 shows full-pipeline autoresearch achieving $3\times$ higher efficiency in Cleanup and 37% higher in Gathering than GEPA at matched compute, with run-to-run spread an order of magnitude tighter on the harder fairness objective.

Automated AI research. A small but growing body of work delegates the design of ML pipelines to LLM coding agents. Karpathy’s autoresearch [4] runs a coding agent that modifies `train.py` for nanoGPT pretraining and is rewarded for validation loss. Our system shares this autoresearch architecture (coding agent + frozen evaluation harness + diff-based history) but targets a different inner loop (multi-agent policy synthesis instead of single-model pretraining) and a different objective (multi-agent social welfare instead of validation bits-per-byte). To our knowledge this is the first instantiation of the autoresearch paradigm in a multi-agent decision-making domain.

289 **Automated mechanism and information design.** Classical automated mechanism design [7]
 290 computes optimal allocation rules for self-interested agents under explicit incentive constraints,
 291 while information design [8] studies how a principal commits to a signaling policy that shapes a
 292 receiver’s behavior. Our outer agent solves a related but distinct problem: $\mathcal{M}$ is not strategically
 293 deceptive (it follows instructions) but is *boundedly rational*, so the researcher must decide what
 294 information, helpers, and structure to expose so that $\mathcal{M}$ writes policies achieving the principal’s
 295 welfare goal. Section 4 shows that the pipelines $\mathcal{R}$ produces are genuinely a function of the welfare
 296 objective, supporting this information-design framing empirically.

297 6 Discussion and Conclusion

298 We have presented a two-level framework where an AI coding agent autonomously discovers
 299 pipeline configurations that improve the output of an inner-loop LLM system. Applied to multi-
 300 agent policy synthesis in social dilemmas, the researcher agent reliably exceeds hand-designed base-
 301 lines across 12 independent runs, and converges on qualitatively similar strategies within each game-
 302 objective condition. The framework requires no domain-specific scaffolding: the agent operates on
 303 a standard software repository using the same tools (file editing, shell commands, git) available to a
 304 human researcher.

305 **Mechanism design in action.** Our results empirically support the mechanism design interpreta-
 306 tion of Section 3.4. The researcher acts as an information designer: under Φ_U , it reveals efficiency-
 307 oriented information (waste counts, zone assignments) that guides $\mathcal{M}$ toward productive but unequal
 308 role allocation (Listing 5). Under $\Phi_{\min}$, it additionally reveals fairness-oriented structure such as
 309 rotation schedules and per-agent equity feedback (Listings 3, 4), guiding $\mathcal{M}$ toward egalitarian coordi-
 310 nation. The striking absence of a fairness–efficiency tradeoff with Gemini suggests that in public
 311 goods provision, the mechanism design problem has a cooperative optimum where both objectives
 312 align.

313 **Two types of dilemma.** The Cleanup–Gathering contrast reveals a structural insight about social
 314 dilemmas: *provision* dilemmas (asymmetric costs, one agent’s contribution benefits all) require ex-
 315 plicit fairness mechanisms, while *restraint* dilemmas (symmetric costs, each agent faces the same
 316 temptation) achieve fairness naturally when coordination improves. This distinction, well-known
 317 in the common-pool resource literature, is rediscovered here by an autonomous AI agent, which
 318 suggests it is a robust structural feature of these environments.

319 **Human oversight and audit.** Our system is fully autonomous by design, but its architecture of-
 320 fers natural affordances for human-in-the-loop oversight. Because the researcher $\mathcal{R}$ operates on a
 321 standard git repository, a domain expert can audit the full history of code diffs Δ_j alongside their
 322 metric outcomes $(\Phi_j, \mathbf{m}_j)$, exactly as they would review a colleague’s pull requests. The keep/dis-
 323 card decision (Algorithm 1, line 11) is a lightweight intervention point: a human could override
 324 the accept threshold, veto modifications that introduce undesirable mechanisms (e.g., exploitative
 325 role assignments), or steer the researcher toward under-explored regions of the configuration space
 326 by editing the experiment history. More broadly, the separation between the researcher $\mathcal{R}$ and the
 327 frozen evaluation Φ creates a *delegation boundary*: the human defines *what* to optimize (the welfare
 328 objective), while the agent decides *how*. This mirrors the expert-in-the-loop design patterns identi-
 329 fied in deployment-oriented work on AI agents for discovery, where trust calibration depends on the
 330 legibility of the agent’s decision trail rather than on real-time supervision.

331 **Future work.** Several directions emerge. First, we are interested in applying the two-level frame-
 332 work to other LLM-driven pipelines (code optimization, scientific experiment design, infrastructure
 333 tuning), where the same architecture applies with a different inner loop and objective Φ ; next, ad-
 334 versarial objectives can be intriguing, to test whether the researcher discovers exploitative pipeline
 335 configurations, exposing reward hacking risks; and asymmetric programs: extend to settings where
 336 agents run *different* source code (the present setup is symmetric in code but already supports hetero-
 337 geneous behavior via `agent_id`; relaxing the shared-program constraint enables richer per-agent
 338 specialization and partial-information settings).

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A Additional Results

Table 3: Strategies discovered by the researcher in Cleanup across 4 efficiency-optimized and 4 maximin-optimized independent runs.

Strategy	Φ_U (4 runs)	$\Phi_{\min}$ (4 runs)
Waste-counting helpers	4/4	4/4
Zone/lane partitioning	3/4	4/4
Anti-regression feedback	3/4	2/4
Worked policy examples	2/4	3/4
Cleaning cost economics	2/4	3/4
Time-based role rotation	0/4	4/4

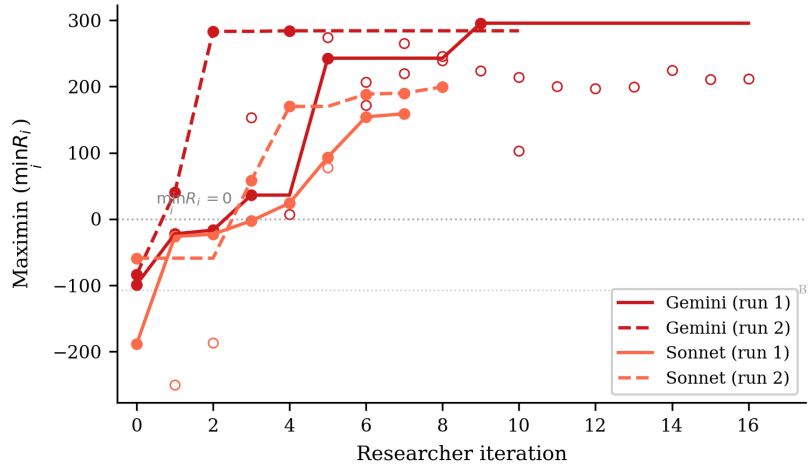


Figure 3: Maximin ($\min_i R_i$) across researcher iterations for the 4 Cleanup maximin-optimized runs. All runs transform deeply negative baselines (worst-off agents losing reward) into substantially positive values. Gemini reaches ~ 290 while Sonnet reaches ~ 160 – 200 . The dashed line marks $\min_i R_i = 0$ (no agent loses reward overall).

B Compute Requirements

Table 4 reports wall-clock time for all 12 autonomous runs and the 6 GEPA comparison runs. *Inner loop* measures total time spent on policy LLM generation and environment simulation across all evaluations; *total* (estimated from run-directory timestamps) additionally includes the researcher agent’s analysis, code editing, and decision-making between evaluations.

Cost breakdown. Each inner loop evaluation involves $K=2$ – 3 policy LLM generation calls (each $\sim 2k$ input tokens, $\sim 1.5k$ output tokens) followed by multi-seed simulation (5–12 seeds, ~ 15 – $30s$ total). Policy LLM generation dominates inner loop time (86–97%), with Sonnet evaluations taking $\sim 2\times$ longer than Gemini due to extended thinking. The researcher agent (Claude Opus 4.6, running via Claude Code CLI) accounts for 41% of total wall-clock time (25.6h of the 62.2h total across all 12 runs), spent reading results, editing pipeline source files, and planning modifications. In monetary

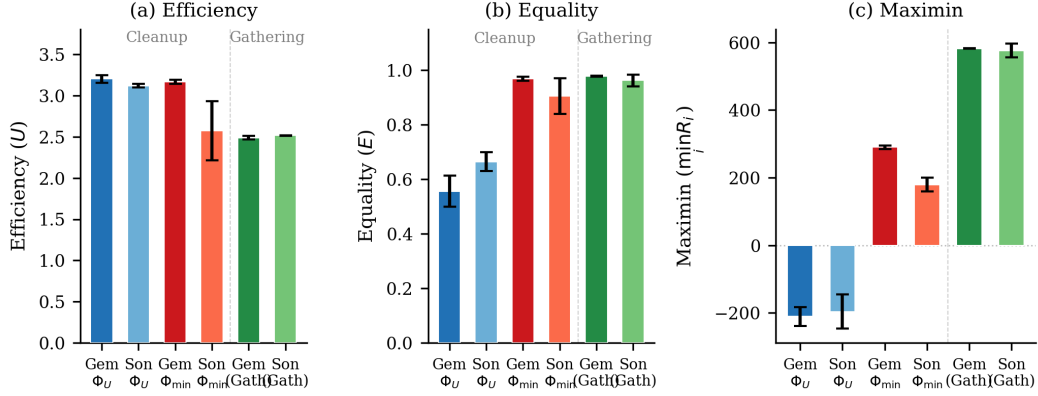


Figure 4: Final metrics across all conditions. (a) Efficiency: all conditions converge to $U \approx 2.5$ – 3.2 . (b) Equality: maximin-optimized Cleanup runs achieve $E \approx 1.0$, while efficiency-optimized runs show $E \approx 0.5$ – 0.7 ; Gathering achieves high equality regardless. (c) Maximin: the sharpest contrast—efficiency optimization leaves the worst-off agent deeply negative ($\min_i R_i \approx -200$), while maximin optimization transforms it to $+290$ (Gemini) or $+179$ (Sonnet). Gathering achieves high maximin (~ 580) under efficiency optimization alone. Error bars show s.d. across runs.

Table 4: Wall-clock time per autonomous run. Evals: total inner loop evaluations including initial baseline. Per eval: mean wall-clock per single evaluation. Total: end-to-end including researcher overhead.

Policy LLM	Φ	Evals	Inner loop (h)	Per eval (min)	Total (h)
<i>Cleanup</i> ($N=10$)					
Gemini	Φ_U	11	2.6	14	3.7
Gemini	Φ_U	18	4.1	14	6.4
Sonnet	Φ_U	4	2.1	32	6.1
Sonnet	Φ_U	7	5.0	43	6.1
Gemini	$\Phi_{\min}$	17	3.0	11	4.3
Gemini	$\Phi_{\min}$	11	3.6	20	5.0
Sonnet	$\Phi_{\min}$	8	4.5	33	8.8
Sonnet	$\Phi_{\min}$	9	5.3	36	13.2
<i>Gathering</i> ($N=4$)					
Sonnet	Φ_U	3	1.7	33	2.2
Gemini	Φ_U	5	1.4	17	1.9
Gemini	Φ_U	3	0.9	19	1.1
Sonnet	Φ_U	4	2.3	35	3.4
All 12 runs		100	36.6	22	62.2
GEPA (6 runs)		6	5.0	50	—

terms, the researcher agent is the dominant cost: each outer iteration consumes ~ 50 – 100 k context tokens in the Opus session, whereas each inner loop evaluation uses only ~ 5 – 10 k policy LLM tokens. The 6 GEPA comparison runs required 5.0h of compute with no researcher agent, operating at lower cost per run but achieving substantially lower performance (Table 2).

C Smaller Open-Weight Model: Gemma 4 26B

We test the framework with Gemma 4 26B-A4B-IT (Google), a 26B-parameter open-weight model substantially smaller than the frontier models in the main experiments. We run three experiments—two on Cleanup ($N=10$) optimizing efficiency and maximin respectively, and one on Gathering ($N=4$) optimizing efficiency—all with Opus 4.6 as the researcher $\mathcal{R}$.

Setup. In all three experiments, Gemma starts from complete failure: the baseline pipeline produces $U=-10.0$ on Cleanup (agents CLEAN-spam without collecting apples) and $U=0.0$ on Gathering (broken BFS calls), compared to $U=1.93/0.86$ (Gemini/Sonnet on Cleanup) and $U=2.04/0.03$ (Gemini/Sonnet on Gathering). The researcher ran $J=10-18$ outer iterations per condition.

Results. Table 5 presents the best configurations discovered. Under efficiency optimization, the researcher achieves $U=0.87$ —a dramatic recovery from the broken baseline but far below frontier models ($U\approx 3.2$). The researcher compensates for the model’s limited capability by reducing inner-loop iterations to $K=1$ (avoiding the regression that plagued $K\geq 2$ runs) and providing highly structured worked examples with explicit cleaning-role assignment.

Table 5: Autoresearch with Gemma 4 26B-A4B-IT. Single run per condition. Baseline: pipeline from [3].

Game	Target	U	E	$\min_i R_i$	Keep rate
Cleanup	Baseline	-10.0	1.00 [†]	-1000	—
	Φ_U	0.87	-1.11	-371	6/19
	$\Phi_{\min}$	1.71	0.94	137	9/17
Gathering	Baseline	0.0	1.00 [†]	0	—
	Φ_U	2.44	0.98	580	4/11

[†]Equality is trivially 1.0 because all agents receive near-zero reward.

Maximin optimization rescues efficiency. Strikingly, under maximin optimization the researcher achieves *higher* efficiency ($U=1.71$) than under direct efficiency optimization ($U=0.87$), alongside strong equality ($E=0.94$) and positive worst-off welfare ($\min_i R_i=137$). This reversal—absent in frontier models, where both objectives yield $U\approx 3.1-3.2$ —occurs because the maximin objective forces the researcher toward coordination mechanisms (rotating cleaning duties, index-based apple assignment) that simultaneously improve collective output. Under efficiency-only optimization, the researcher converges on a local optimum—static role assignment with 25% dedicated cleaners—that a 26B model can implement but that caps performance well below the frontier.

This suggests that *for models below a capability threshold, fairness objectives may serve as better optimization targets for overall social welfare than direct efficiency maximization*. The structured coordination enforced by the maximin objective provides a scaffold that compensates for the weaker model’s difficulty implementing complex strategies from strategic hints alone.

Gathering: full recovery on a simpler game. On Gathering ($N=4$), the researcher brings Gemma from $U=0.0$ to $U=2.44$ with $E=0.98$ and $\min_i R_i=580$, after 10 outer iterations (4 kept). These values nearly match frontier models (Gemini: $U=2.49$, Sonnet: $U=2.52$; Table 2). The researcher discovers the same Voronoi partitioning and respawn-aware camping strategies found in the main experiments, and increases inner-loop iterations to $K=5$ to allow sufficient refinement. The contrast with Cleanup—where Gemma peaks at $U=0.87/1.71$ versus frontier $U\approx 3.2$ —suggests that the researcher can fully compensate for model weakness on simpler coordination tasks (4 agents, symmetric costs) but not on harder provision dilemmas (10 agents, asymmetric costs).

D Researcher-Authored Pipeline Artifacts

The previous appendix shows the policies π_c^* that the inner loop *outputs*; this one shows the configuration $c = (p, \phi, \mathcal{H}, \iota)$ (Section 4, Table 1) that the researcher $\mathcal{R}$ *authored* to produce them. Excerpts are taken from the final commit on the dedicated git branch of the corresponding run; we reproduce the prose verbatim, with author commentary in [brackets] and elisions marked We organize by artifact type so each subsection makes a within-type contrast (e.g., Φ_U vs. $\Phi_{\min}$, or Cleanup vs. Gathering).

D.1 Helper library $\mathcal{H}$: coordination primitives

The researcher adds primitives that the policy LLM can call as black boxes, sparing it from re-implementing tricky logic on every iteration. Two patterns dominate: (i) *state inspection* (waste

counts, fractions, beam-yield scoring) and (ii) *coordination primitives* (zone assignment, role rotation). We show one of each.

Listing 1: Cleanup, $\Phi_{\min}$ – band-based apple zoning helper added by the researcher in exp5. This is the primitive that lets the policy in Listing 6 keep collectors within their own row band, preventing all 7 gatherers from racing to the same apple.

```

455 1 def get_my_apples(env, agent_id):
456 2     """Alive apples in this agent's horizontal row band.
457 3     Divides the orchard into n_agents bands; falls back to all apples if empty."""
458 4     aid = int(agent_id)
459 5     n = int(env.n_agents)
460 6     band_h = env.height / n
461 7     band_lo, band_hi = aid * band_h, (aid + 1) * band_h
462 8
463 9     my_apples, all_apples = set(), set()
464 10    for i in range(env.n_apples):
465 11        if env.apple_alive[i]:
466 12            ar = int(env._apple_pos[i, 0]); ac = int(env._apple_pos[i, 1])
467 13            all_apples.add((ar, ac))
468 14            if band_lo <= ar < band_hi:
469 15                my_apples.add((ar, ac))
470 16    return my_apples if my_apples else all_apples
471

```

Listing 2: Gathering – BFS-Voronoi territory + respawn-aware waiting helpers added by the researcher in gather-exp3. These are the primitives that Listing 8 composes into a one-line policy: “go to my_zone_apples, otherwise nearest_respawning_apple.”

```

473 1 def voronoi_zones(env):
474 2     """Multi-source BFS Voronoi over walkable cells.
475 3     Returns (row, col) -> agent_id; ties broken by lower agent_id."""
476 4     queue, visited = deque(), {}
477 5     for a_id in range(env.n_agents):
478 6         if int(env.agent_timeout[a_id]) == 0:
479 7             ar = int(env.agent_pos[a_id][0]); ac = int(env.agent_pos[a_id][1])
480 8             queue.append((ar, ac, a_id))
481 9             visited[(ar, ac)] = a_id
482 10    while queue:
483 11        r, c, a_id = queue.popleft()
484 12        for dr, dc in [(-1,0), (1,0), (0,-1), (0,1)]:
485 13            nr, nc = r + dr, c + dc
486 14            if 0 <= nr < env.height and 0 <= nc < env.width \
487 15                and not env.walls[nr, nc] and (nr, nc) not in visited:
488 16                visited[(nr, nc)] = a_id
489 17                queue.append((nr, nc, a_id))
490 18    return visited
491
492 19
493 20 def nearest_respawning_apple(env, agent_id, zones=None, max_timer=10):
494 21     """Nearest dead apple in MY zone respawning within max_timer steps.
495 22     Returns (row, col) of best wait spot, or None."""
496 23     if zones is None: zones = voronoi_zones(env)
497 24     best_pos, best_t, best_d = None, max_timer + 1, float('inf')
498 25     ar = int(env.agent_pos[agent_id][0]); ac = int(env.agent_pos[agent_id][1])
499 26     for i in range(env.n_apples):
500 27         if not env.apple_alive[i]:
501 28             pos = (int(env._apple_pos[i][0]), int(env._apple_pos[i][1]))
502 29             if zones.get(pos) == agent_id:
503 30                 t = int(env.apple_timer[i])
504 31                 if t <= max_timer:
505 32                     d = abs(pos[0] - ar) + abs(pos[1] - ac)
506 33                     if t < best_t or (t == best_t and d < best_d):
507 34                         best_pos, best_t, best_d = pos, t, d
508 35    return best_pos
509

```

The choice of helper depends on the dilemma’s geometry: row-bands suffice when fairness is the main concern (Cleanup maximin) but full BFS-Voronoi is needed when wall-aware territory ownership matters (Gathering). The researcher does not fix this in advance — it picks the simpler primitive whenever it works.

514 D.2 System prompt p : from neutral framing to strategic briefing

The unmodified system prompt is a neutral API description: “Write a policy that maximizes per-agent reward.” Under maximin optimization, the researcher rewrites the prompt into a multi-section strategic briefing. The excerpt below shows the pieces it added to the Cleanup prompt in exp5; the full prompt grew from 165 to 325 lines.

Listing 3: Cleanup, $\Phi_{\min}$ – excerpts from the researcher-rewritten system prompt (exp5). The unmodified baseline contained only API documentation; everything below was added by $\mathcal{R}$.

```

519 ## CRITICAL OBJECTIVE: Rawlsian Fairness (Maximin)
520
521 Your goal is to maximize the minimum per-agent total return across all
522 agents. This is the "maximin" or Rawlsian welfare criterion.
523
524 This means: it is NOT enough for the *average* agent to do well. The
525 worst-off agent must do as well as possible. A policy where 8 agents
526 each earn +200 but 2 agents each earn -100 is TERRIBLE (maximin = -100).
527
528 Key implication: cleaning costs (-1 per CLEAN action) must be shared
529 equitably among ALL agents. If you assign fixed "cleaner" roles, those
530 agents will accumulate large negative rewards and destroy your maximin score.
531
532 ## Strategy for Maximin: ROLE ROTATION + APPLE ZONING
533
534 The optimal strategy for maximin involves:
535 1. Shared cleaning duty: ALL agents take turns cleaning using a rotation
536    schedule based on 'agent_id' and 'env._step_count'. For example:
537    'is_my_cleaning_turn = (agent_id + env._step_count // SHIFT) % env.n_agents < NUM_CLEANERS'
538    where SHIFT is ~50 steps and NUM_CLEANERS is 2-3.
539 2. When NOT your turn: collect apples from YOUR ZONE using
540    'get_my_apples(env, agent_id)' -- prevents all gatherers competing
541    for the same nearest apple.
542 3. NEVER use BEAM (action 6): it costs -1 and causes -50 to the target.
543 4. Keep waste density low: aim for ~2-3 active cleaners at any time.
544
545 [... API documentation, helper documentation, working example ...]
546
547 IMPORTANT:
548 - NEVER assign permanent cleaning roles to specific agents -- this kills maximin.
549 - Use env._step_count for time-based role rotation so all agents share cleaning duty.
550 - NEVER use BEAM (action 6) -- it destroys both agents' rewards.
551

```

Two things stand out. First, the researcher writes the formula it expects $\mathcal{M}$ to use almost verbatim — $(\text{agent_id} + \text{env}._step_count // \text{SHIFT}) \% \text{env}._n_agents < \text{NUM_CLEANERS}$ — and the policy in Listing 6 uses essentially this template. Second, the researcher *teaches by counter-example*: the explicit “8 agents earn +200, 2 earn -100, maximin = -100” makes the failure mode of Φ_U -style static roles (Listing 5) concretely visible to $\mathcal{M}$. This is the information-design move predicted by the mechanism-design framing in Section 3.4.

The Gathering prompt evolves along a different axis (no maximin, no rotation). Its researcher-added “Key Strategic Insights” section instead emphasizes (i) *self-play implies never beam* and (ii) *Voronoi partition each step*, exactly matching what the policy in Listing 8 implements.

562 D.3 Feedback ϕ : adaptive diagnostics

The baseline feedback function from [3] shows reward + four social metrics with definitions and stops there. The researcher turns it into a state machine: it inspects the latest metrics and conditionally injects different hints. Two different runs produced two qualitatively different diagnostics — the same metric ($\bar{r}_k$ or $\min_i R_i$) is repurposed to either *stabilize* a working policy or *redirect* a failing one.

Listing 4: Adaptive feedback diagnostics. *Top*: efficiency-optimized run (exp1) – a stability guard prevents iter-3 regression once U is high. *Bottom*: maximin-optimized run (exp5) – two fairness diagnostics that fire when the worst-off agent loses reward.

```

568 1 # --- Cleanup, Phi_U feedback (exp1, pipeline/feedback.py final) ---
569 2 last_eff = history[-1]["metrics"]["efficiency"]
570 3 if last_eff >= 2.5:
571 4     parts.append(
572 5         """CRITICAL -- DO NOT REGRESS*: The current policy achieves high "
573 6         "efficiency (>=2.5). You MUST output a policy that is nearly identical "
574 7         "to the current one. Copy the current policy code and make AT MOST "
575 8         "one small targeted improvement (e.g., adjust a single numeric "
576 9         "threshold). If you are not confident a change will help, output the "
577 10        "current policy UNCHANGED. A regression here means the run fails."
578 11
579 12 # --- Cleanup, Phi_min feedback (exp5, pipeline/feedback.py final) ---
580 13 last_maximin = history[-1]["metrics"].get("maximin", 0)
581 14 last_avg = history[-1]["reward_avg"]
582 15 if last_maximin < 0:
583

```

```

584 16 parts.append(
585 17     f"**FAIRNESS ALERT**: maximin={last_maximin:.1f} is NEGATIVE. "
586 18     f"The worst-off agent lost reward overall while average was {last_avg:.1f}. "
587 19     "This means cleaning duties are NOT shared equitably. "
588 20     "Use time-based role rotation (env._step_count) so ALL agents share "
589 21     "cleaning costs equally. NEVER assign permanent cleaner roles.")
590 22 elif last_maximin < last_avg * 0.5:
591 23     parts.append(
592 24         f"**FAIRNESS WARNING**: maximin={last_maximin:.1f} is much lower than "
593 25         f"average={last_avg:.1f}. The gap suggests unequal cleaning burden. "
594 26         "Ensure ALL agents rotate through cleaning duty.")

```

The two diagnostics are written by independent researcher runs but converge on a common pattern: *thresholded interventions on a primary metric*. They are also a common cause of the low run-to-run variance reported in Finding 1: when a working policy lands in the high- U basin, the stability guard prevents the LLM from over-refining and tipping out of it (the $K=3$ regressions visible in Section 4’s “Common failure modes” (4)), and when an unfair policy lands in the negative-maximin region, the alert injects exactly the rotation hint that recovers it.

D.4 Iteration logic ι : per-condition hyperparameters

Although ι has the smallest source surface, the researcher’s edits to it explain a meaningful fraction of the variance reduction noted in Finding 1. Table 6 summarizes the values shipped in the final commit of each best-performing run.

Table 6: Iteration-logic (ι) values in the final commit of selected runs. K = inner-loop iterations, $|S|$ = evaluation seeds, “thinking” = extended-thinking token budget passed to $\mathcal{M}$. Baseline ([3]) is $K=3$, $|S|=5$, thinking 16k.

Run	Game / objective	K	$ S $	thinking	Researcher’s stated rationale
exp1 (Gemini)	Cleanup, Φ_U	3	5	16k	default; stability comes from feedback hint
exp4 (Sonnet)	Cleanup, Φ_U	3	5	32k	“Sonnet was over-refining at 16k thinking”
exp5 (Gemini)	Cleanup, $\Phi_{\min}$	2	12	16k	“ $K=2$ avoids iter-3 regression; 12 seeds for variance control”
exp7 (Sonnet)	Cleanup, $\Phi_{\min}$	3	5	10k	“reduced thinking from 16k – Sonnet was generating runaway code at 16k”
gather-exp3 (Gemini)	Gathering, Φ_U	3	5	16k	default; Gathering converges in $K \leq 3$

The maximin Gemini run is the most informative case: the researcher discovered that with $K=3$ and $|S|=5$, identical configurations could yield $\min_i R_i=295$ on one set of seeds and $\min_i R_i=100$ on another (this is the “variance exposure” failure mode of Section 4’s common failures). It then walked $|S|$ from $5 \rightarrow 8 \rightarrow 12$ over three outer iterations until the seed-averaged signal was stable enough that the policy LLM stopped chasing noise. The policy in Listing 6 is sampled at $|S|=12$.

D.5 Cross-artifact observations

- *Helpers do the algebra; prompts pick the strategy class.* The researcher uses helpers (1, 2) to put non-trivial primitives at $\mathcal{M}$ ’s fingertips, and the prompt (3) to tell $\mathcal{M}$ *which* primitive to compose. Neither is sufficient alone — prompts without supporting helpers produced policies that “mention rotation” but failed to implement it; helpers without prompt updates often went unused.
- *Feedback is the only adaptive component.* p , $\mathcal{H}$, and ι are static within a run; ϕ is the only place where the researcher gets to change what $\mathcal{M}$ sees *during* the inner loop, and it uses thresholded diagnostics (Listing 4) to do so. This is what mostly drives the run-to-run variance reduction.
- *The same artifact has different content under each objective.* The Cleanup prompt under Φ_U omits “rotation” and “maximin” entirely; the same prompt under $\Phi_{\min}$ devotes its entire opening section to it. The configuration c is genuinely a function of the welfare objective Φ , not a fixed pipeline parameterized by it — consistent with the information-design framing of Section 3.4.
- *The researcher’s edits are short and targeted.* The full diff between the baseline pipeline and the best maximin Gemini configuration totals ~ 300 added lines spread over four files; ~ 80 of them appear in the listings above. The remaining content is API documentation and worked examples, included for completeness but not strategically novel.

627 E Selected Generated Policies

628 The code excerpts below are verbatim outputs of the policy synthesizer $\mathcal{M}$, taken from the final
 629 inner-loop iteration of the best-performing run in each condition. To fit the page, we elide repeated
 630 boilerplate (BFS scaffolding, fallback branches) with `...` and add commentary in [brackets];
 631 nothing else has been edited. The listings illustrate the qualitative differences highlighted in Findings
 632 2–4 of Section 4: *static* vs. *rotating* cleaning roles in Cleanup, and the *Voronoi* + *respawn-timer*
 633 structure in Gathering. All four policies were independently generated by the LLM under different
 634 seeds, models, and outer-loop runs.

635 E.1 Cleanup, Φ_U (Gemini): static interleaved roles + dynamic threshold

636 Best run: $U=3.25$, $E=0.61$, $\min_i R_i = -182$. The synthesizer assigns *permanent* cleaning roles
 637 via `agent_id mod  $n < \tau$` , where the threshold τ depends smoothly on the current waste fraction.
 638 Higher pollution recruits more cleaners, but the *identity* of each cleaner is fixed — agents $0-\tau-1$
 639 always pay the cost. This drives U to its ceiling but leaves equality at 0.61 and pushes the worst-off
 640 agent to $\min_i R_i = -182$.

Listing 5: Cleanup, Φ_U – Gemini 3.1 Pro. Static role assignment with waste-tiered threshold (verbatim, abridged).

```

641 1 def policy(env, agent_id) -> int:
642 2     if int(env.agent_timeout[agent_id]) > 0:
643 3         return 7 # STAND while removed
644 4
645 5     waste_frac = waste_fraction(env)
646 6     n_agents = env.n_agents
647 7
648 8     # [Smooth dynamic threshold: more pollution -> more cleaners]
649 9     if waste_frac > 0.35: threshold_cleaners = n_agents
650 10    elif waste_frac > 0.25: threshold_cleaners = int(n_agents * 0.7)
651 11    elif waste_frac > 0.15: threshold_cleaners = int(n_agents * 0.5)
652 12    elif waste_frac > 0.05: threshold_cleaners = int(n_agents * 0.3)
653 13    else: threshold_cleaners = max(1, int(n_agents * 0.2))
654 14
655 15    # [STATIC role: cleaners are ALWAYS the lowest-id agents -- no rotation]
656 16    is_cleaner = (agent_id % n_agents) < threshold_cleaners
657 17
658 18    if is_cleaner:
659 19        best_orient, n_waste = find_best_clean_orientation(env, agent_id)
660 20        if n_waste >= 1:
661 21            cur_orient = int(env.agent_orient[agent_id])
662 22            if best_orient == cur_orient: return 8 # CLEAN
663 23            if (cur_orient + 1) % 4 == best_orient: return 5 # ROTATE_RIGHT
664 24            elif (cur_orient - 1) % 4 == best_orient: return 4 # ROTATE_LEFT
665 25            else: return 4
666 26
667 27        # [Competitor-aware BFS toward waste in own row band]
668 28        c_idx = agent_id % n_agents
669 29        target_r = int(((c_idx + 0.5) / threshold_cleaners) * env.height)
670 30        # ... pick waste cell minimizing dist + |row - target_r| + 50 *
671 31        # closer_cleaners
672 32        # ... bfs_toward and direction_to_action
673 33    else:
674 34        # GATHERER: row-banded sweep, competitor-aware
675 35        # assigned_row = ((g_idx + 0.5) / n_gatherers) * env.height
676 36        # ... pick alive apple minimizing dist + |row - assigned_row| + 50 *
677 37        # closer_gatherers
678 38    ...

```

680 E.2 Cleanup, $\Phi_{\min}$ (Gemini): time-rotated, geographically interleaved roles

681 Best run: $U=3.19$, $E=0.98$, $\min_i R_i = 296$. Switching the outer objective from U to $\min_i R_i$ flips
 682 the role-assignment scheme entirely: `role_idx` now depends on `agent_id + step/50`, so every
 683 agent rotates through cleaner and gatherer duty in a 50-step cycle. The cleaner indices $\{1, 4, 8\}$ are
 684 spatially *interleaved* along the river (rather than contiguous) so that whoever is on duty has a short
 685 walk to their assigned river slice. This single change accounts for the $E=0.55 \rightarrow 0.98$ jump and the
 686 $\min_i R_i = -182 \rightarrow 296$ swing reported in Table 2, with negligible efficiency loss.

Listing 6: Cleanup, $\Phi_{\min}$ – Gemini 3.1 Pro. Time-rotated roles with interleaved geography (verbatim, abridged).

```

687 1 def policy(env, agent_id) -> int:
688 2     if int(env.agent_timeout[agent_id]) > 0:
689 3         return 7 # STAND while removed
690 4
691 5     aid = int(agent_id)
692 6     step = int(env._step_count)
693 7     n = int(env.n_agents)
694 8
695 9     # [TIME ROTATION: every 50 steps each agent's role index advances]
696 10    if n == 10:
697 11        role_idx = (aid + step // 50) % 10
698 12        # [Cleaner indices {1,4,8} are SPATIALLY INTERLEAVED, not contiguous,
699 13        # so each on-duty cleaner has a short walk to its river slice]
700 14        if role_idx == 1: role_type, zone_idx = 'C', 0
701 15        elif role_idx == 4: role_type, zone_idx = 'C', 1
702 16        elif role_idx == 8: role_type, zone_idx = 'C', 2
703 17        else:
704 18            role_type = 'G'
705 19            g_map = {0:0, 2:1, 3:2, 5:3, 6:4, 7:5, 9:6}
706 20            zone_idx = g_map[role_idx]
707 21            num_cleaners, num_gatherers = 3, 7
708 22
709 23    if role_type == 'C':
710 24        # [Each on-duty cleaner owns one horizontal slice of the river]
711 25        row_min = int( zone_idx * env.height / num_cleaners)
712 26        row_max = int((zone_idx + 1) * env.height / num_cleaners) - 1
713 27
714 28        # [Score every (row, col, orientation) in the slice by beam yield;
715 29        # fire if facing >=2 waste, else move to a >=2-yield position]
716 30        best_positions, good_positions = set(), set()
717 31        for r in range(row_min, row_max + 1):
718 32            for c in range(10):
719 33                if env.walls[r, c] or env.waste[r, c]: continue
720 34                y_max = max(get_clean_yield(env, r, c, o) for o in range(4))
721 35                if y_max >= 2: best_positions.add((r, c))
722 36                if y_max >= 1: good_positions.add((r, c))
723 37            # ... rotate / CLEAN / bfs_to_target_set as appropriate
724 38        else:
725 39            # GATHERER: collect apples ONLY inside the rotating row band
726 40            # row_min/row_max for num_gatherers slices ...
727 41            ...
728 41

```

730 E.3 Cleanup, $\Phi_{\min}$ (Sonnet): independent rediscovery of duty rotation

731 Best run: $U=2.93$, $E=0.83$, $\min_i R_i=154$. Run with a different policy LLM (Sonnet 4.6) and on
732 a different outer-loop trajectory, the synthesizer arrives at the *same* structural insight as the Gemini
733 maximin run: a phase counter ($\text{agent_id} + \text{step}/50$) mod n rotates which 2 of the 10 agents clean
734 at any given time, and a *separate* 200-step zone counter rotates which 5-row band each collector
735 sweeps. The two rotation periods (50 and 200 steps) ensure every agent visits every cleaning slot
736 *and* every apple zone within one episode — a structural fairness invariant the researcher discovered
737 without any explicit instruction to “rotate.”

Listing 7: Cleanup, $\Phi_{\min}$ – Sonnet 4.6. Phase-rotated cleaning + zone-rotated collection (verbatim, abridged).

```

738 1 def policy(env, agent_id) -> int:
739 2     if int(env.agent_timeout[agent_id]) > 0:
740 3         return 7 # STAND while removed
741 4
742 5     n = env.n_agents
743 6     step = env._step_count
744 7     wf = waste_fraction(env)
745 8
746 9     # --- ROLE ASSIGNMENT: fair 2-cleaner rotation ---
747 10    # [50-step phases x 20 phases / episode -> each agent cleans 4 phases (~20%)]
748 11    phase = step // 50
749 12    cleaner_rank = (agent_id + phase) % n # [cycles 0..9 fairly]
750 13    is_cleaner = cleaner_rank < 2 # [exactly 2 cleaners per phase]
751 14
752 15    # [Emergency override: all agents clean when waste crosses the spawn cliff]
753 16    in_emergency = wf >= 0.40
754 17    if in_emergency: is_cleaner = True
755 18
756 19    # [River split: cleaner_slot 0 -> top half, 1 -> bottom half]
757 20    cleaner_slot = (agent_id % 2) if in_emergency else cleaner_rank
758 21

```

```

760 22     if is_cleaner and wf > 0.04:
761 23         # [Scan 4 orientations from current position; if best beam shot
762 24         # covers >=3 waste cells, fire; else step to a strictly better neighbor]
763 25         # ... (best_o, best_cnt) = max over 4 directions
764 26         # ... if best_cnt >= 3: rotate to best_o, then CLEAN
765 27         # ... else: scan 4 adjacent cells x 4 orientations for higher yield
766 28         # ... if no waste in beam range: bfs to waste_set in OWN river half,
767 29         # fall back to whole river
768 30
769 31     # --- COLLECTOR LOGIC: 5-zone rotating apple collection ---
770 32     # [5 zones x 200 steps = 1000 steps -> each agent visits every zone once;
771 33     # 2 agents share each zone at any time, reducing competition 10 -> 2]
772 34     bounds = zone_boundaries(env, 5)
773 35     zone_phase = step // 200
774 36     zone = (agent_id + zone_phase) % 5
775 37     z_start, z_end = bounds[zone], bounds[zone + 1]
776 38
777 39     zone_apples = {(int(env._apple_pos[i][0]), int(env._apple_pos[i][1]))
778 40                    for i in range(env.n_apples) if env.apple_alive[i]
779 41                    and z_start <= int(env._apple_pos[i][0]) < z_end}
780 42     # ... bfs_to_target_set(zone_apples), else bfs_nearest_apple as fallback
781 43     return 7

```

783 E.4 Gathering (Gemini): wall-aware Voronoi + spatiotemporal targeting

784 Best run: $U=2.47$, $E=0.98$, $\min_i R_i=580$. In Gathering, where costs are symmetric, no role differentiation is needed: the entire optimization is spatial and temporal. The synthesizer (i) partitions cells into BFS-Voronoi territories owned by individual agents, (ii) runs a single centralized BFS from its own position to score every reachable cell with exact wall-aware distances, and (iii) for both alive and respawning apples in its territory, computes the *earliest collection time* $\max(\text{walk_dist}, \text{respawn_timer})$ and targets the minimum. When the agent must wait on a dead apple, it steps onto a *non-spawn* adjacent cell rather than blocking a respawn point. This single sort key — $\max(\text{distance}, \text{timer})$ — subsumes both “go to nearest apple” and “camp respawn” as special cases.

Listing 8: Gathering – Gemini 3.1 Pro. Voronoi territories + spatiotemporal priority targeting (verbatim, abridged).

```

793 1 def policy(env, agent_id) -> int:
794 2     if int(env.agent_timeout[agent_id]) > 0:
795 3         return 7
796 4
797 5
798 6     r, c = int(env.agent_pos[agent_id][0]), int(env.agent_pos[agent_id][1])
799 7     orient = int(env.agent_orient[agent_id])
800 8
801 9     # [Wall-aware Voronoi over the gridworld; spawn_points = all apple cells]
802 10    zones = voronoi_zones(env)
803 11    spawn_points = {(int(p[0]), int(p[1])) for p in env._apple_pos}
804 12
805 13    # [Single centralized BFS: exact distance + first move to every reachable cell]
806 14    distances, first_moves = {}, {}
807 15    q = deque([(r, c, 0, None)])
808 16    visited = {(r, c)}
809 17    while q:
810 18        cr, cc, d, fm = q.popleft()
811 19        distances[(cr, cc)] = d
812 20        first_moves[(cr, cc)] = fm
813 21        for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
814 22            nr, nc = cr + dr, cc + dc
815 23            if 0 <= nr < env.height and 0 <= nc < env.width and not env.walls[nr, nc]:
816 24                if (nr, nc) not in visited:
817 25                    visited.add((nr, nc))
818 26                    q.append((nr, nc, d + 1, fm if fm is not None else (dr, dc)))
819 27
820 28    # [Both alive AND respawning apples in MY zone, with their respawn timer]
821 29    my_zone_spawns = []
822 30    for i in range(env.n_apples):
823 31        pr, pc = int(env._apple_pos[i][0]), int(env._apple_pos[i][1])
824 32        if zones.get((pr, pc)) == agent_id and (pr, pc) in distances:
825 33            my_zone_spawns.append((pr, pc, int(env.apple_timer[i]), distances[(pr, pc)]))
826 34
827 35    # [SPATIOTEMPORAL PRIORITY: earliest collection time first.
828 36    # score = max(walk_distance, respawn_timer); ties broken by sooner timer]
829 37    my_zone_spawns.sort(key=lambda x: (max(x[3], x[2]), x[2], x[3]))

```

```

832 37
833 38     for pr, pc, timer, dist in my_zone_spawns:
834 39         if timer == 0: # [Alive apple: walk straight to it]
835 40             if dist == 0: return 7
836 41             fm = first_moves[(pr, pc)]
837 42             return direction_to_action(fm[0], fm[1], orient)
838 43         else: # [Dead apple: camp on safe adjacent cell]
839 44             # ... pick nearest neighbor (nr,nc) that is NOT in spawn_points,
840 45             #         navigate there and STAND while waiting for respawn
841 46             ...
842 47         # [Global poaching fallback if zone is empty: nearest alive apple anywhere]
843 48         ...

```

845 E.5 Cross-condition observations

846 Several patterns are visible across these four listings (and confirmed by inspecting the remaining 8
847 runs not shown):

- 848 • *Role assignment encodes the welfare objective.* Static `agent_id <  $\tau$` (Listing 5) maximizes
849 U but harms $\min_i R_i$. Time-rotated `(agent_id + step//T) % n` (Listings 6, 7) maximizes
850 $\min_i R_i$ at $\leq 1\%$ efficiency cost (Gemini).
- 851 • *Convergent rediscovery across LLMs.* Gemini and Sonnet, on independent runs, both arrive at
852 the `(id + phase) % n` duty-rotation idiom with similar phase lengths ($T \approx 50$ steps) under
853 maximin — and both omit it under efficiency.
- 854 • *Game structure dictates strategy class.* Cleanup policies are dominated by role-assignment logic
855 (cleaner vs. gatherer); Gathering policies are dominated by territory partitioning and respawn-
856 timer reasoning, with no role differentiation at all. The researcher does not need to be told which
857 class of strategy to write.
- 858 • *Earliest-collection-time as a unifying score.* The Gathering policy’s `max(walk, timer)` key (List-
859 ing 8) is structurally identical across the 4 Gathering runs (both Gemini and Sonnet), differing
860 only in how the Voronoi diagram is computed — Manhattan approximation, fully BFS-based, or
861 cached helper.